CAPSTONE SYSTEMS SYNTHESIS

Module map

- Module 16: Capstone Systems Synthesis
- Connected lab: lab-16_capstone-systems-synthesis.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Biology across scales

Mechanism chains

Capstone explanation and revision

Module 16

Feedback and constraint

Evidence from multiple modules

lab-16_capstone-systems-synthesis.md

CAPSTONE SYSTEMS SYNTHESIS

Learning objectives

- Connect concepts from at least three modules in one explanation.
- Trace a mechanism across biological scales.
- Identify feedback loops, constraints, and tradeoffs.
- Match evidence types to claims at different scales.
- Revise a systems explanation using new evidence.

OBJECTIVE 1
Connect concepts from at least three modules in one explanation.

OBJECTIVE 2
Trace a mechanism across biological scales.

OBJECTIVE 3
Identify feedback loops, constraints, and tradeoffs.

OBJECTIVE 4
Match evidence types to claims at different scales.

OBJECTIVE 5
Revise a systems explanation using new evidence.

CAPSTONE SYSTEMS SYNTHESIS

Topic sequence

- Biology across scales: Biological explanations become stronger when they connect molecules, cells, organisms, populations, and ecosystems.
- Mechanism chains: Mechanism chains show how one change can produce effects across scales.
- Feedback and constraint: Feedback loops and constraints explain why biological systems do not change without limits.
- Evidence from multiple modules: Evidence from labs, models, and observations must match the scale of the claim.
- Capstone explanation and revision: Capstone reasoning combines concepts into a clear claim that can be tested and revised.

TOPIC 1
Biology across scales

TOPIC 2
Mechanism chains

TOPIC 3
Feedback and constraint

TOPIC 4
Evidence from multiple modules

TOPIC 5
Capstone explanation and revision

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Evidence Map

- Module focus: Capstone Systems Synthesis
- Central claim: Capstone Systems Synthesis explains cross-scale biological explanation through mechanisms, evidence chains, and revision.
- Key nodes to connect: Biology across scales, Mechanism chains, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-16_capstone-systems-synthesis.md supplies an evidence surface for the map.

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Reasoning Sequence

- Module focus: Capstone Systems Synthesis
- Trace the model: Biology across scales -> Mechanism chains -> Feedback and constraint -> Evidence from multiple modules
- Inputs become outputs: Biology across scales, Mechanism chains -> Connect concepts from at least three modules in one explanation., Trace a mechanism across biological scales.
- Feedback check: lab-16_capstone-systems-synthesis.md evidence checks whether students can explain feedback and constraint.

CAPSTONE SYSTEMS SYNTHESIS

Terms as evidence handles

- System: Interacting parts whose relationships shape outcomes.
- Scale: Level of biological organization.
- Mechanism: Causal process that explains how something happens.
- Feedback loop: Circular causal relationship that amplifies or reduces change.
- Constraint: Limit that shapes what a system can do.
- Tradeoff: Benefit in one function paired with cost elsewhere.

SYSTEM
Interacting parts whose relationships shape outcomes.

SCALE
Level of biological organization.

MECHANISM
Causal process that explains how something happens.

FEEDBACK LOOP
Circular causal relationship that amplifies or reduces change.

CONSTRAINT
Limit that shapes what a system can do.

TRADEOFF
Benefit in one function paired with cost elsewhere.

CAPSTONE SYSTEMS SYNTHESIS

Lab connection

- Lab file: lab-16_capstone-systems-synthesis.md
- Lab evidence should test or illustrate: Feedback and constraint
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Biology across scales

EVIDENCE
lab-16_capstone-systems-synthesis.md

CLAIM
Connect concepts from at least three modules in one explanation.

CAPSTONE SYSTEMS SYNTHESIS

Contrast check

- Do not stop at: Biological explanations become stronger when they connect molecules, cells, organisms, populations, and ecosystems.
- Stronger explanation: Capstone reasoning combines concepts into a clear claim that can be tested and revised.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Biological explanations become stronger when they connect molecules, cells, organisms, populations, and ecosystems.

DEEPER

Capstone reasoning combines concepts into a clear claim that can be tested and revised.

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Retrieval and Lab Check

- Module focus: Capstone Systems Synthesis
- Retrieval prompt: How can a molecular change affect an ecosystem?
- Answer check: Connect concepts from at least three modules in one explanation.
- Required terms: System, Scale, Mechanism
- Lab connection: lab-16_capstone-systems-synthesis.md; lab-16_capstone-systems-synthesis.md supplies evidence for capstone synthesis evidence from lab-to-course integration.

CAPSTONE SYSTEMS SYNTHESIS

Practice quiz bridge

- 1. Which statement best defines System in Module 16?
- 2. Which learning goal best supports the topic "Mechanism chains"?
- 3. How should lab-16_capstone-systems-synthesis.md support Module 16?
- 4. Which retrieval move best prepares a student for Capstone Systems Synthesis?

QUIZ 1
A

QUIZ 2
B

QUIZ 3
C

QUIZ 4
D

CAPSTONE SYSTEMS SYNTHESIS

Synthesis and exit ticket

- Exit claim: explain biology across scales using evidence from lab-16_capstone-systems-synthesis.md.
- Must include: System, Scale, Mechanism.
- Revision prompt: what evidence would change your explanation?

CLAIM
Biology across scales

EVIDENCE
lab-16_capstone-systems-synthesis.md

REVISION
What would change your mind?